

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-7446
Project Name	ComicCraft - AI Comic Story Creator
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Image generation is slow: each comic needs 5 sequential image generations.	Long waiting time per comic; limits how many users can be served at once.	High
2	No user accounts or saved comic library.	Users cannot revisit, edit or share earlier comics.	Medium
3	Depends on external Gemini API limits and a local GPU for Stable Diffusion.	Generation may fail or slow down due to quotas or missing hardware.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local FastAPI server serving one user at a time.	Deploy on cloud with load balancing and a background task queue (e.g. Celery/Redis) to serve many users.
2	Data Storage	Images and PDFs saved in local static folders.	Move files to cloud storage (e.g. AWS S3) and add a database for user comics and history.
3	Performance	Sequential image generation for each panel.	Use GPU instances, cache results and generate panel images in parallel or with faster models.
4	Security	API keys kept in .env file; no user login.	Add user authentication, HTTPS, rate limiting, input validation and secure secret management.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	User accounts and a personal comic library.	1-2 months	Users can save, revisit and edit their comics.
Phase 3	Cloud deployment with GPU and async task queue.	3-4 months	Faster generation and support for many concurrent users.
Phase 4	Multi-page story arcs, speech bubbles and sharing.	5-6 months	Richer comics with more engagement and easy sharing.